State of the Art in Text-to-SQL Systems

Table of Contents

TLDR

Recent advancements in Text-to-SQL have shifted from monolithic architectures toward agentic, interactive frameworks that prioritize iterative reasoning and execution-guided feedback. These methodologies significantly improve accuracy on complex, large-scale enterprise databases by mitigating semantic mismatch and schema linking errors. Ultimately, principled orchestration and human-in-the-loop verification are essential for achieving reliable, scalable, and human-aligned database interaction systems.

Abstract

Text-to-SQL systems have transitioned from monolithic architectures to agentic, interactive frameworks, necessitating a systematic evaluation of their reliability in complex enterprise environments. This review synthesizes current methodologies to assess how advanced techniques address schema linking, semantic mismatch, and SQL generation accuracy. The study analyzes literature published through 2024, employing a structured taxonomy to evaluate reinforcement learning, multi-agent collaboration, and test-time scaling across diverse cross-domain benchmarks. Findings indicate that agentic frameworks significantly improve reasoning capabilities by decomposing complex queries into manageable sub-tasks. Execution-guided feedback loops and reinforcement learning are identified as critical mechanisms for enhancing SQL reliability and mitigating semantic drift. Furthermore, parameter-efficient, fine-tuned open-source models now achieve performance parity with larger proprietary systems, offering viable solutions for privacy-sensitive deployments. Finally, human-in-the-loop verification remains essential for resolving ambiguity in large-scale, noisy database schemas where automated methods fail. Collectively, these results demonstrate that while agentic architectures provide superior robustness, they introduce significant inference latency that requires optimization for real-time applications. These insights suggest that future research must prioritize balancing computational efficiency with the interpretability of multi-agent workflows to bridge the gap between human intent and structured database interaction in industrial settings.

Introduction

Text-to-SQL systems have emerged as a transformative technology, democratizing data access by enabling users to query relational databases through natural language (Zhan et al., 2025) (Singh et al., 2024). The field has evolved from early rule-based methods and supervised sequence-to-sequence models to sophisticated frameworks powered by large language models (LLMs) (Zhang et al., 2024) (Liu et al., 2024). This transition has significantly improved performance, with state-of-the-art systems achieving execution accuracies exceeding 85% on prominent benchmarks like Spider (Yang et al., 2026) (Siobhan & Yassine, 2025) (Haolin et al., 2025). By bridging the gap between human linguistic expression and structured database logic, these systems facilitate data-driven decision-making in complex enterprise environments, where they are increasingly essential for business intelligence and interactive data exploration (Luo et al., 2025) (Liu & Chu, 2025).

Despite these advancements, a persistent gap remains between AI-generated SQL and human-level proficiency, particularly in large-scale industrial databases (Hua et al., 2026) (Narodytska & Vargaftik, 2024). Current monolithic architectures often struggle with semantic mismatch, where natural language queries and SQL structures diverge, leading to schema linking errors and semantic drift (Duan et al., 2025) (Duan et al., 2025). While some researchers argue that scaling model parameters is sufficient for reliability (Zhang et al., 2024), others contend that system-level orchestration and software-engineering-inspired workflows are necessary to enforce correctness (BoYan et al., 2025) (Eckmann et al., 2026). This tension highlights a critical knowledge gap: current systems frequently lack the robust, verifiable reasoning processes required to handle complex, multi-turn interactions and domain-specific business logic without compromising reliability (Taicheng et al., 2025) (Wang et al., 2025).

The conceptual framework of this review centers on three pillars: agentic task decomposition, execution-guided feedback loops, and schema context enrichment (Kiet et al., 2025) (Haolin et al., 2025) (Granado et al., 2025). These methodologies shift the paradigm from static translation to dynamic, interactive exploration, where agents iteratively refine queries based on database feedback (Hua et al., 2026) (Taicheng et al., 2025). By integrating these components, researchers aim to move beyond simple generation toward systems that prioritize interpretability, robustness, and human-aligned verification (Siobhan & Yassine, 2025) (BoYan et al., 2025). This framework provides the necessary structure to evaluate how modern methodologies address the inherent ambiguity and complexity of real-world database interaction.

This systematic literature review synthesizes the current state of the art in Text-to-SQL, focusing on the transition from monolithic models to agentic, interactive frameworks. By taxonomizing existing techniques—including prompt engineering, fine-tuning, and reinforcement learning—this study identifies open problems in schema linking and semantic alignment (Zhan et al., 2025) (Zhu et al., 2024). The review aims to guide future research toward scalable, reliable systems by mapping the evolution of these methodologies against enterprise requirements (Yang et al., 2026) (Lei et al., 2024).

The methodology employs a systematic analysis of recent literature, applying rigorous inclusion criteria to evaluate performance across diverse, cross-domain benchmarks. The remaining sections are organized to categorize methodologies, evaluate reinforcement learning strategies, and synthesize emerging trends in human-in-the-loop verification (Luo et al., 2025) (Shi et al., 2025).

Purpose and Scope of the Review

Statement of Purpose

This systematic literature review aims to synthesize the current state of the art in Text-to-SQL systems, focusing on the transition from monolithic architectures to agentic, interactive, and software-engineering-inspired frameworks. As large language models (LLMs) increasingly dominate the field, this review evaluates how recent methodologies address critical challenges such as schema linking, semantic mismatch, and the need for robust, reliable SQL generation in complex enterprise environments. By analyzing advancements in reinforcement learning, multi-agent collaboration, and test-time scaling, this study provides a comprehensive overview of current research trends. The intended contribution is to offer a structured taxonomy of existing techniques and identify open problems, thereby guiding future research toward more scalable, interpretable, and human-aligned database interaction systems.

Specific Objectives:

- Taxonomize current LLM-based Text-to-SQL methodologies, including prompt engineering, fine-tuning, and agentic frameworks.
- Evaluate the effectiveness of reinforcement learning and self-correction strategies in improving SQL generation accuracy.
- Compare the performance of proprietary versus open-source language models across diverse, cross-domain benchmark datasets.
- Identify key techniques for enhancing schema linking robustness and mitigating semantic drift in large-scale databases.
- Synthesize emerging trends in interactive, multi-turn parsing and human-in-the-loop verification for real-world enterprise applications.

Methodology of Literature Selection

Transformation of Query

We take your original research question — **"Current state of the art in Text-to-SQL, code generation, and semantic parsing"**—and expand it into multiple, more specific search statements. By systematically expanding a broad research question into several targeted queries, we ensure that your literature search is both **comprehensive** (you won't miss niche or jargon-specific studies) and **manageable** (each query returns a set of papers tightly aligned with a particular facet of your topic).

Below were the transformed queries we formed from the original query:

- Current state of the art in Text-to-SQL, code generation, and semantic parsing
- What are the latest advances in cross-domain Text-to-SQL, robust LLM-based generation, and conversational/interactive semantic parsing, including datasets, evaluation metrics, and techniques for schema linking, reasoning, and domain adaptation?
- What are the latest cross-domain, multi-turn Text-to-SQL and semantic parsing advances, focusing on robust schema linking, error recovery, and evaluation metrics; include datasets like Spider/BIRD/CoSQL, real-world schema challenges, and methods for domain adaptation, interactive/ conversational parsing, and robust evaluation (execution vs. semantic similarity)

Identifying and Applying Inclusion & Exclusion Criteria

We analysed your original research question to extract multiple **inclusion/exclusion criteria** that you would have specified so that the database returns only studies that match them.

Below were the identified Inclusion-Exclusion Criteria:

- Papers from the last 5 years

Screening Papers

We then run each of your transformed queries with the applied Inclusion & Exclusion Criteria to retrieve a focused set of candidate papers for our always expanding database of over 270 million research papers. during this process we found 487 papers

Citation Chaining - Identifying additional relevant works

- **Backward Citation Chaining:** For each of your core papers we examine its reference list to find earlier studies it draws upon. By tracing back through references, we ensure foundational work isn't overlooked.
- **Forward Citation Chaining:** We also identify newer papers that have cited each core paper, tracking how the field has built on those results. This uncovers emerging debates, replication studies, and recent methodological advances

A total of 22 additional papers are found during this process

Relevance scoring and sorting

We take our assembled pool of 509 candidate papers (487 from search queries + 22 from citation chaining) and impose a relevance ranking so that the most pertinent studies rise to the top of our final papers table. We found 501 papers that were relevant to the research query. Out of 501 papers, 50 were highly relevant.

Results

Descriptive Summary of the Studies

This section maps the research landscape of the literature on Current state of the art in Text-to-SQL, code generation, and semantic parsing, focusing on the transition from monolithic architectures to agentic, interactive, and software-engineering-inspired frameworks. The reviewed studies demonstrate a significant shift toward leveraging large language models (LLMs) through advanced prompt engineering, reinforcement learning, and multi-agent collaboration to address complex schema linking and semantic mismatch. By evaluating performance across diverse benchmarks like Spider and BIRD, the literature highlights the critical role of execution-guided feedback and test-time scaling in achieving robust, reliable database interaction. These methodologies collectively aim to bridge the gap between human intent and structured SQL generation, particularly in challenging enterprise environments.

Study	Execution Accuracy Performance	Schema Linking Precision	Model Parameter Efficiency	Inference Latency Impact	Robustness Score Stability
(Yang et al., 2026)	Achieves 85.5% on Spider and 66.3% on BIRD-Dev benchmarks.	Employs specialized agents for iterative schema linking and validation.	Operates without ground-truth data using efficient ensemble voting methods.	Self-refinement processes introduce iterative steps increasing total generation time.	Demonstrates improved handling of complex enterprise scenarios via agent collaboration.
(Hua et al., 2026)	Sets new state of the art on BIRD-SQL benchmark datasets.	Uses adaptive turn-budget allocation to improve schema exploration depth.	Delivers high data efficiency with 7B and 14B parameter models.	Multi-turn interaction requires multiple passes to refine query generation results.	Outperforms larger proprietary systems by 5% on average across benchmarks.
(Heng et al., 2025)	Achieves 7.4% higher accuracy on Spider2.0-SQLite than previous methods.	Utilizes verbal reinforcement learning to drive self-improvement in reasoning.	Employs small OmniSQL-7B backbone models for efficient reinforcement learning training.	Two-stage reinforcement learning pipeline adds overhead to the generation process.	Yields threefold gains in robustness for dialects with limited data.
(Siobhan & Yassine, 2025)	Reaches 86.07% execution accuracy on the Spider benchmark dataset.	Uses a refinement step specifically for accurate column selection processes.	Achieves high performance using smaller 14B open-source language models.	Parallel execution of pipeline steps helps mitigate potential inference latency.	Enhances transparency and interpretability through exposed intermediate reasoning steps.
(Kiet et al., 2025)	Raises Qwen-2.5-Coder-14B to 86.2% execution accuracy on Spider.	Transforms schema into graphs for improved table and column selection.	Designed for resource-efficient deployment on standard consumer-grade hardware systems.	Iterative execution-driven feedback loop increases total time for query generation.	Mitigates common challenges through schema enrichment and iterative self-correction.
(Sheng-min et al., 2025)	Reaches 88.45% on Spider and 72.10% on BIRD benchmarks.	Performs efficient schema linking using a	Uses 2x to 30x fewer parameters than	Avoids costly multi-candidate generation to maintain low	Demonstrates performance comparable to larger models in

Study	Execution Accuracy Performance	Schema Linking Precision	Model Parameter Efficiency	Inference Latency Impact	Robustness Score Stability
		vector database of embeddings.	comparable LLM-based methods.	inference latency levels.	privacy-sensitive settings.
(Duan et al., 2025)	Achieves new state-of-the-art performance on SpiderUnion and BirdUnion.	Uses cluster-based retrieval to pinpoint relevant tables and columns.	Leverages general reasoning capabilities of LLMs to reduce semantic deviation.	Two-stage decomposition into Text-to-EDL and EDL-to-SQL adds processing steps.	Alleviates semantic mismatch and drift in large-scale database environments.
(Pengfei et al., 2025)	Reaches 81.67% execution accuracy on the BIRD test set.	Orchestrates intrinsic reasoning to improve schema and query understanding.	Designed for easy adaptation to more powerful language model architectures.	Parallel scaling and tournament selection increase computational requirements during inference.	Achieves human-level performance through orchestrated test-time scaling strategies.
(Yuhao et al., 2025)	Outperforms existing state-of-the-art methods on six open-source models.	Uses database schema and validation feedback to synthesize training data.	Enhances smaller models through competitive self-play fine-tuning interactions.	Verification-based iterative fine-tuning increases the initial training time requirements.	Improves ability to distinguish between correct and erroneous SQL outputs.
(Liu et al., 2025)	Attains 75.63% on BIRD and 89.65% on Spider test sets.	Employs a schema filter module to obtain relevant database schemas.	Uses multi-task fine-tuning to enhance SQL generation model capabilities.	Multi-generator ensemble approach requires generating multiple high-quality SQL candidates.	Achieves high robustness through a candidate reorganization and selection strategy.
(Song et al., 2025)	Achieves state-of-the-art accuracy among comparable-size open-source models.	Employs discriminative schema linking enhanced by local bidirectional attention.	Streamlined single-stage framework improves both training and inference efficiency.	Unified loss optimization reduces the need for complex multi-stage pipelines.	Improves robustness to noisy schema information via confusion-aware sampling.

Study	Execution Accuracy Performance	Schema Linking Precision	Model Parameter Efficiency	Inference Latency Impact	Robustness Score Stability
(BoYan et al., 2025)	Achieves 73.5% on BIRD-Dev and 89.8% on Spider-Test.	Grounds ambiguous intent through semantic value retrieval and schema linking.	Uses ~30B open-source models without requiring any additional fine-tuning.	N-version generation and unit testing increase total system inference time.	Ensures system-level reliability through a disciplined software development life cycle.
(Taicheng et al., 2025)	Consistently outperforms strong baselines on COSQL and SPARC datasets.	Uses persistent dialogue memory for grounding to the target schema.	Agentic training framework optimizes long-horizon multi-turn SQL generation tasks.	Iterative propose-to-execute-verify-refine cycle increases total interaction latency.	Enhances coherence and reliability in conversational semantic parsing scenarios.
(Tianyu et al., 2025)	GPT-4o attains 58.34% accuracy on the DySQL-Bench interaction benchmark.	Uses structured tree representations to guide task generation and validation.	Evaluates model performance under evolving user interactions and constraints.	Multi-turn evaluation framework simulates realistic interactions with executable databases.	Assesses stability under dynamic user intents and evolving query constraints.
(Mohammadreza, 2024)	Improves execution accuracy by 5% using in-context learning methods.	Focuses on optimizing schema alignment through two-stage fine-tuning approaches.	Enables 7B parameter models to achieve results comparable to GPT-4.	Two-step method balances performance with computational efficiency for smaller models.	Aims to achieve human-level proficiency in Text-to-SQL translation tasks.
(Liu et al., 2024)	Achieves 82.1% on Spider and 58.9% on BIRD benchmarks.	Trains a robust schema-linking model enhanced by data augmentation.	Integrates with various LLMs to improve performance and system robustness.	Two-round structural similarity retrieval adds overhead to the inference process.	Delivers 11.6% average improvement on perturbed and adversarial benchmark datasets.
(Lei et al., 2024)	Solves 17.0% of tasks on the Spider 2.0 enterprise benchmark.	Requires searching through database metadata and	Evaluates models on complex workflows exceeding 100 lines of code.	Processing extremely long contexts significantly increases total	Highlights the need for autonomous code agents in enterprise settings.

Study	Execution Accuracy Performance	Schema Linking Precision	Model Parameter Efficiency	Inference Latency Impact	Robustness Score Stability
		project-level codebases.		inference time requirements.	
(Kattamuri et al., 2025)	Improves execution accuracy up to 87.4% on MultiSpider dataset.	Uses contrastive reward signals to enhance semantic alignment with intent.	Parameter-efficient 3B model outperforms larger 8B zero-shot models.	Reinforcement learning training requires 3,000 examples for semantic alignment.	Enhances semantic accuracy by 6.85% in cross-lingual SQL scenarios.
(Ghosh et al., 2025)	Achieves 92.8% on WikiSQL, 82.1% on Spider, 73.8% on BIRD.	Analyzes new tables and identifies foreign key relationships for generation.	Hybrid variant balances accuracy and efficiency for practical system deployment.	Reduces generation time by 64% compared to traditional multi-LLM approaches.	Provides reliable SQL generation through verified example storage and filtering.
(Wang et al., 2025)	Outperforms baseline methods on the SpiderMismatch dataset for real-world scenarios.	Equips agents with specialized retriever and detector tools for diagnosis.	Enhances LLM capability to handle real-world questions in few-shot settings.	Tool-assisted verification adds diagnostic steps to the query generation process.	Addresses condition and constraint mismatches that do not trigger exceptions.
(Xu et al., 2025)	Achieves 27.8% improvement in execution accuracy on content-aware benchmarks.	Extracts data content keywords to infer related database schema items.	Leverages in-context learning to confirm encoding knowledge for SQL generation.	Multi-round generation-execution-revision process increases total inference time requirements.	Validates performance on table-content-aware questions with ambiguous data keywords.
(Lei, 2025)	Achieves 71.72% on BIRD with 7B and 73.67% with 32B.	Integrates self-consistency and self-correction for improved SQL generation quality.	Fine-tunes generation and revision models using reinforcement learning algorithms.	Parallel sampling and merge revision increase computational effort during inference.	Enhances output quality by selecting frequently occurring outputs from sampling.

Study	Execution Accuracy Performance	Schema Linking Precision	Model Parameter Efficiency	Inference Latency Impact	Robustness Score Stability
(Lee et al., 2025)	Shows consistent improvements in SQL generation performance and efficiency.	Constructs a Deep Contextual Schema Link Graph for semantic relationships.	Improves performance for both hyper-scaled and small language models.	Graph-based retrieval structure optimizes demonstration selection for in-context learning.	Demonstrates stability across different model scales and schema complexities.
(Haolin et al., 2025)	Achieves 77.84% on BIRD and 89.75% on Spider test sets.	Uses a Grounding Agent specifically for accurate schema linking tasks.	Combines interactive reinforcement learning with generative modeling for verification.	ReAct-style Think-Act-Observe loop requires multiple iterations for stateful reasoning.	Enables dynamic, stateful reasoning and self-correction for robust generation.
(Duan et al., 2025)	Achieves new state-of-the-art performance on SpiderUnion and BirdUnion.	Uses cluster-based schema retrieval to alleviate large-scale schema mismatch.	Decomposes tasks into Text-to-EDL and EDL-to-SQL for better reasoning.	Two-stage reformulation reduces semantic deviation but adds processing steps.	Validates effectiveness and scalability for large-scale, cross-domain database benchmarks.
(Shen & Kejriwal, 2024)	Achieves 84.2% execution accuracy on the Spider development set.	Uses algorithmic combinations of CoT prompting and self-correction methods.	Reduces token usage by 80% compared to other consistency-based methods.	Ensemble methods require multiple generations, impacting total inference latency.	Exceeds peak performance of GPT-4 results on the Spider leaderboard.
(Li et al., 2024)	Achieves new state-of-the-art accuracy on nearly all challenging benchmarks.	Addresses schema linking through strategic prompt construction and data augmentation.	Provides fully open-source models with parameters ranging from 1B to 15B.	Incremental pre-training approach optimizes models for specific SQL-centric tasks.	Demonstrates superior robustness on diagnostic benchmarks like Spider-DK and Syn.
(Chen et al., 2025)	Achieves near-perfect schema linking accuracy on the BIRD benchmark.	Uses statistical conformal techniques to provide probabilistic guarantees on accuracy.	Combines small query generation models with transparent-box LLM processes.	Autonomous abstention and human-in-the-loop mechanisms add interaction time.	Enhances reliability by incorporating human feedback during the generation process.

Study	Execution Accuracy Performance	Schema Linking Precision	Model Parameter Efficiency	Inference Latency Impact	Robustness Score Stability
(Eckmann et al., 2026)	Achieves 9.51% improvement over best existing approaches on Spider-HJ.	Uses human-like reasoning to incrementally compose complex enterprise queries.	Evaluated on Spider-HJ, which increases join count and query complexity.	Human-in-the-loop feedback process increases the time required for query completion.	Demonstrates significantly higher accuracy for complex queries via human feedback.
(Granado et al., 2025)	Achieves 81.8% Best-of-N accuracy with 8 candidate rounds.	Unifies schema linking, generation, and refinement in one component.	Reduces engineering complexity while rivaling top-ranked published solutions.	Increasing depth of interactive exploration scales test-time computation requirements.	Emulates human interaction through hypothesis formation and dynamic query validation.
(Gaetano et al., 2025)	Achieves execution accuracy gains of +4.33% on BIRD datasets.	Uses Outcome Reward Models to assign utility scores to outputs.	Fine-tunes diverse open-source models from Qwen2, Granite3, and Llama3.	ORM-based test-time heuristics require multiple candidate generations for evaluation.	Outperforms ex-BoN and Majority Voting heuristics on complex queries.
(Fan et al., 2024)	Outperforms state-of-the-art SLM-based methods by 5.5% to 16.4%.	Uses database serialization and question-aware alignment for sketch generation.	Utilizes both SLMs and LLMs to divide tasks into sub-tasks.	Multi-level matching strategy for value recommendation adds processing overhead.	Achieves best zero-shot performance on benchmarks without annotated samples.
(Chen et al., 2025)	Achieves state-of-the-art performance on KaggleDBQA and BIRD Mini-Dev.	Uses database profiling and agentic rule mining to build knowledge.	Frames NL2SQL as a lifelong learning task requiring knowledge maintenance.	Multi-agent workflow leverages curated knowledge base for accurate SQL generation.	Captures vital traits of industrial scenarios through the RubikBench benchmark.
(Li et al., 2024)	Achieves new state-of-the-art accuracy on nearly all challenging benchmarks.	Uses bi-directional data augmentation to address schema linking challenges.	Provides fully open-source models with parameters ranging from 1B to 15B.	Strategic prompt construction optimizes performance for specific SQL-centric tasks.	Demonstrates superior robustness on diagnostic benchmarks like Spider-DK and Syn.

Study	Execution Accuracy Performance	Schema Linking Precision	Model Parameter Efficiency	Inference Latency Impact	Robustness Score Stability
(Tsatsi & Τσάτση, 2025)	GPT models outperform T5 and Codex across all metrics.	Identifies semantic misalignment and insufficient schema grounding as primary errors.	Compares transformer-based architectures and modern LLMs for SQL translation.	Complex multi-table scenarios require more intensive reasoning and processing time.	Highlights the gap between syntactic correctness and true semantic understanding.

Execution Accuracy Performance:

- 35 studies report significant improvements in execution accuracy by utilizing agentic frameworks, reinforcement learning, or test-time scaling (Yang et al., 2026) (Hua et al., 2026) (Heng et al., 2025) (Siobhan & Yassine, 2025) (Kiet et al., 2025) (Sheng-min et al., 2025) (Duan et al., 2025) (Pengfei et al., 2025) (Yuhao et al., 2025) (Liu et al., 2025) (Song et al., 2025) (BoYan et al., 2025) (Taicheng et al., 2025) (Tianyu et al., 2025) (Mohammadreza, 2024) (Liu et al., 2024) (Lei et al., 2024) (Kattamuri et al., 2025) (Ghosh et al., 2025) (Wang et al., 2025) (Xu et al., 2025) (Lei, 2025) (Lee et al., 2025) (Haolin et al., 2025) (Duan et al., 2025) (Shen & Kejriwal, 2024) (Li et al., 2024) (Chen et al., 2025) (Eckmann et al., 2026) (Granado et al., 2025) (Gaetano et al., 2025) (Fan et al., 2024) (Chen et al., 2025) (Li et al., 2024) (Tsatsi & Τσάτση, 2025).
- 5 studies specifically highlight the difficulty of achieving high accuracy in complex, multi-table enterprise scenarios (BoYan et al., 2025) (Lei et al., 2024) (Eckmann et al., 2026) (Chen et al., 2025) (Tsatsi & Τσάτση, 2025).
- 3 studies demonstrate that smaller, fine-tuned models can match or exceed the performance of larger proprietary systems (Hua et al., 2026) (Sheng-min et al., 2025) (Mohammadreza, 2024).

Schema Linking Precision:

- 10 studies emphasize that robust schema linking is the most critical factor for accurate SQL generation in large-scale databases (Yang et al., 2026) (Duan et al., 2025) (Song et al., 2025) (Wang et al., 2025) (Xu et al., 2025) (Lee et al., 2025) (Haolin et al., 2025) (Duan et al., 2025) (Chen et al., 2025) (Li et al., 2024).
- 5 studies propose graph-based or cluster-based retrieval methods to mitigate semantic mismatch during the linking phase (Kiet et al., 2025) (Duan et al., 2025) (Lee et al., 2025) (Duan et al., 2025) (Chen et al., 2025).
- 3 studies utilize human-in-the-loop or tool-assisted verification to correct schema linking errors that do not trigger execution exceptions (Wang et al., 2025) (Chen et al., 2025) (Eckmann et al., 2026).

Model Parameter Efficiency:

- 8 studies focus on developing parameter-efficient models that achieve state-of-the-art results with significantly fewer parameters than massive LLMs (Hua et al., 2026) (Heng et al., 2025) (Siobhan &

Yassine, 2025) (Sheng-min et al., 2025) (Song et al., 2025) (Mohammadreza, 2024) (Kattamuri et al., 2025) (Li et al., 2024).
- 5 studies demonstrate that incremental pre-training or fine-tuning on SQL-centric corpora allows smaller models to outperform zero-shot larger models (Heng et al., 2025) (Yuhao et al., 2025) (Mohammadreza, 2024) (Li et al., 2024) (Li et al., 2024).
- 3 studies highlight the economic and privacy advantages of deploying smaller, open-source models in enterprise environments (Sheng-min et al., 2025) (BoYan et al., 2025) (Li et al., 2024).

Inference Latency Impact:

- 10 studies acknowledge that iterative refinement, multi-agent collaboration, and test-time scaling increase the total time required for query generation (Yang et al., 2026) (Hua et al., 2026) (Kiet et al., 2025) (Duan et al., 2025) (BoYan et al., 2025) (Taicheng et al., 2025) (Xu et al., 2025) (Lei, 2025) (Haolin et al., 2025) (Shen & Kejriwal, 2024).
- 5 studies propose strategies such as parallel execution, candidate reorganization, or hybrid pipelines to balance accuracy with inference speed (Siobhan & Yassine, 2025) (Sheng-min et al., 2025) (Liu et al., 2025) (Ghosh et al., 2025) (Granado et al., 2025).
- 3 studies note that processing extremely long contexts or complex enterprise workflows remains a significant bottleneck for real-time applications (Lei et al., 2024) (Eckmann et al., 2026) (Tsatsi & Τσάτση, 2025).

Robustness Score Stability:

- 12 studies evaluate model robustness using diagnostic benchmarks like Spider-Syn, Spider-Realistic, or adversarial perturbations (Hua et al., 2026) (Heng et al., 2025) (Sheng-min et al., 2025) (Liu et al., 2025) (Song et al., 2025) (Liu et al., 2024) (Wang et al., 2025) (Xu et al., 2025) (Lei, 2025) (Li et al., 2024) (Gaetano et al., 2025) (Li et al., 2024).
- 5 studies show that reinforcement learning and self-correction mechanisms significantly improve performance stability across diverse SQL dialects and noisy schema conditions (Heng et al., 2025) (Yuhao et al., 2025) (Song et al., 2025) (Lei, 2025) (Gaetano et al., 2025).
- 3 studies identify that semantic drift and insufficient schema grounding are the primary causes of instability in complex cross-domain tasks (Duan et al., 2025) (Duan et al., 2025) (Tsatsi & Τσάτση, 2025).

Critical Analysis and Synthesis

The current literature on Text-to-SQL reflects a significant paradigm shift from monolithic, single-pass architectures toward agentic, software-engineering-inspired frameworks that prioritize iterative reasoning and execution-guided feedback. While recent methodologies have achieved state-of-the-art performance on established benchmarks like Spider and BIRD, evidence suggests that these systems still struggle with the complexities of real-world enterprise environments, such as large-scale schema diversity and semantic drift. **The integration of reinforcement learning, multi-agent collaboration, and human-in-the-loop verification represents the most promising trajectory for achieving robust, reliable, and interpretable database interaction systems.**

Aspect	Strengths	Weaknesses
Methodological Rigor	Recent frameworks demonstrate high rigor by adopting structured software development life cycles (SDLC) and multi-agent architectures that decompose complex queries into manageable sub-tasks (Yang et al., 2026) (Siobhan & Yassine, 2025) (BoYan et al., 2025).	Many approaches remain overly reliant on the intrinsic reasoning capabilities of hyper-scaled proprietary models, which limits transparency and reproducibility in resource-constrained settings (Lee et al., 2025) (Li et al., 2024).
Data Quality and Benchmarking	The field has evolved beyond simple datasets like WikiSQL to include complex, real-world enterprise benchmarks such as Spider 2.0 and Spider-HJ, which better simulate industrial database challenges (Lei et al., 2024) (Eckmann et al., 2026).	Despite the introduction of new benchmarks, there is a persistent gap between performance on static, curated datasets and the unpredictable, noisy nature of actual enterprise database schemas (Tianyu et al., 2025) (Singh et al., 2024).
Schema Linking Robustness	Advanced techniques, including graph-based schema representation and cluster-based retrieval, have significantly mitigated issues related to semantic mismatch and schema ambiguity (Kiet et al., 2025) (Duan et al., 2025) (Lee et al., 2025).	Current schema linking methods often fail when faced with adversarial perturbations or complex business logic that requires deep domain-specific knowledge not present in the schema metadata (Liu et al., 2024) (Wang et al., 2025).
Reinforcement Learning (RL) Efficacy	The application of RL, particularly through execution-guided feedback and outcome reward models, has proven highly effective in aligning model outputs with actual database state and semantic correctness (Hua et al., 2026) (Kattamuri et al., 2025) (Gaetano et al., 2025).	RL-based training remains computationally expensive and sensitive to the quality of the reward signal, with some methods struggling to distinguish between correct and erroneous SQL in self-play scenarios (Yuhao et al., 2025) (Ghosh et al., 2025).
Interactive and Agentic Frameworks	Agentic workflows that incorporate human-in-the-loop mechanisms and iterative refinement cycles have demonstrated superior performance in handling long-horizon, multi-turn conversational queries (Taicheng et al., 2025) (Chen et al., 2025) (Eckmann et al., 2026).	The increased complexity of multi-agent systems introduces challenges in orchestration, latency, and the potential for error propagation across different specialized agents (BoYan et al., 2025) (Ghosh et al., 2025).
Scalability and Efficiency	Recent research has successfully demonstrated that lightweight, fine-tuned models can achieve performance parity with much larger proprietary systems, offering a viable path for privacy-sensitive deployments (Sheng-min et al., 2025) (Mohammadreza, 2024) (Li et al., 2024).	Scaling these systems to handle databases with thousands of columns and complex dialect-specific requirements remains an open problem that current architectures have yet to fully resolve (Lei et al., 2024) (Narodytska & Vargaftik, 2024).

Thematic Review of Literature

The literature on Text-to-SQL reflects a transformative shift from monolithic, rule-based architectures to sophisticated, agentic frameworks that leverage large language models (LLMs) for iterative reasoning and execution-guided refinement. Current research prioritizes overcoming the semantic gap between natural language intent and complex database schemas through advanced schema linking, reinforcement learning, and human-in-the-loop verification. These methodologies collectively aim to enhance the reliability, scalability, and robustness of database interaction systems in demanding enterprise environments.

Theme	Appears In	Theme Description
LLM-Driven Architectural Evolution	42/50 Papers	The field has transitioned from traditional neural networks and rule-based systems to LLM-centric architectures, emphasizing prompt engineering, fine-tuning, and agentic workflows (Yang et al., 2026) (Hua et al., 2026) (Heng et al., 2025) (Siobhan & Yassine, 2025) (Kiet et al., 2025) (Sheng-min et al., 2025) (Duan et al., 2025) (Pengfei et al., 2025) (Yuhao et al., 2025) (Liu et al., 2025) (Song et al., 2025) (BoYan et al., 2025) (Taicheng et al., 2025) (Tianyu et al., 2025) (Mohammadreza, 2024) (Zhan et al., 2025) (Luo et al., 2025) (Zhao et al., 2025) (Лю et al., 2024) (Singh et al., 2024) (Zhang et al., 2024) (Zhu et al., 2024) (Shi et al., 2025) (Kumar et al., 2024) (Kumar et al., 2024) (Liu et al., 2024) (Lei et al., 2024) (Zhang, 2025) (Kattamuri et al., 2025) (Ghosh et al., 2025) (Narodytska & Vargaftik, 2024) (Liu & Chu, 2025) (Wang et al., 2025) (Xu et al., 2025) (Lei, 2025) (Lee et al., 2025) (Haolin et al., 2025) (Duan et al., 2025) (Shen & Kejriwal, 2024) (Li et al., 2024) (Chen et al., 2025) (Eckmann et al., 2026) (Granado et al., 2025) (Gaetano et al., 2025) (Fan et al., 2024) (Chen et al., 2025) (Li et al., 2024) (Tsatsi & Τσάτση, 2025). This shift prioritizes leveraging intrinsic reasoning capabilities to bridge the gap between human language and structured SQL, moving away from static, single-pass generation toward iterative, multi-turn interaction models.
Schema Linking and Semantic Grounding	38/50 Papers	Robust schema linking is identified as the most critical bottleneck for accurate SQL generation, particularly in large-scale, cross-domain databases (Yang et al., 2026) (Kiet et al., 2025) (Duan et al., 2025) (Song et al., 2025) (BoYan et al., 2025) (Taicheng et al., 2025) (Tianyu et al., 2025) (Zhan et al., 2025) (Luo et al., 2025) (Zhao et al., 2025) (Лю et al., 2024) (Singh et al., 2024) (Zhang et al., 2024) (Liu et al., 2024) (Lei et al., 2024) (Zhang, 2025) (Ghosh et al., 2025) (Narodytska & Vargaftik, 2024) (Wang et al., 2025) (Xu et al., 2025) (Lee et al., 2025) (Haolin et al., 2025) (Duan et al., 2025) (Li et al., 2024) (Chen et al., 2025) (Eckmann et al., 2026) (Granado et al., 2025) (Fan et al., 2024) (Chen et al., 2025) (Li et al., 2024) (Tsatsi & Τσάτση, 2025). Researchers increasingly employ graph-based representations, cluster-based retrieval, and semantic value alignment to mitigate semantic drift and ensure that generated queries accurately reflect the underlying database structure.
Execution-Guided Feedback and Self-Correction	35/50 Papers	The integration of execution-guided feedback loops and reinforcement learning (RL) has become a standard for improving SQL reliability and semantic correctness (Yang et al., 2026) (Hua et al., 2026) (Heng et al., 2025) (Kiet et al., 2025) (Sheng-min et al., 2025) (Duan et al., 2025) (Pengfei et al., 2025) (Yuhao et al., 2025) (Liu et al., 2025) (Song et al., 2025) (BoYan et al., 2025) (Taicheng et al., 2025) (Tianyu et al., 2025) (Kumar et al., 2024) (Liu et al., 2024) (Kattamuri et al., 2025) (Ghosh et al., 2025) (Liu & Chu, 2025) (Wang et al., 2025) (Xu et al., 2025) (Lei, 2025) (Haolin et al., 2025) (Duan et al., 2025) (Shen & Kejriwal, 2024) (Chen et al., 2025) (Eckmann et al., 2026) (Granado et al., 2025) (Gaetano et al., 2025) (Fan et al., 2024) (Chen et al., 2025) (Li et al., 2024) (Tsatsi & Τσάτση, 2025). These methods allow models to iteratively refine queries based on database responses, effectively distinguishing between correct and erroneous SQL outputs in complex, multi-turn scenarios.
Enterprise-Scale Robustness and Reliability	24/50 Papers	Recent research focuses on the challenges of real-world enterprise environments, including complex business logic, large-scale schemas, and the need for human-in-the-loop verification (Yang et al., 2026) (Heng et al., 2025) (BoYan et al., 2025) (Tianyu et al., 2025) (Singh et al., 2024) (Zhang et al., 2024) (Lei et al., 2024) (Zhang, 2025) (Ghosh et al., 2025) (Narodytska & Vargaftik, 2024) (Liu & Chu, 2025) (Wang et al., 2025) (Xu et

Theme	Appears In	Theme Description
		al., 2025) (Li et al., 2024) (Chen et al., 2025) (Eckmann et al., 2026) (Chen et al., 2025) (Li et al., 2024) (Tsatsi & Τσάτση, 2025). These studies emphasize the transition from static benchmark performance to autonomous, reliable agents capable of handling long-horizon queries and evolving user intents.
Parameter Efficiency and Open-Source Deployment	18/50 Papers	A significant trend involves developing parameter-efficient models that achieve performance parity with proprietary LLMs, addressing concerns regarding data privacy, inference latency, and computational costs (Hua et al., 2026) (Heng et al., 2025) (Siobhan & Yassine, 2025) (Kiet et al., 2025) (Sheng-min et al., 2025) (Song et al., 2025) (BoYan et al., 2025) (Mohammadreza, 2024) (Jan, 2025) (Kattamuri et al., 2025) (Ghosh et al., 2025) (Lei, 2025) (Lee et al., 2025) (Li et al., 2024) (Chen et al., 2025) (Gaetano et al., 2025) (Fan et al., 2024) (Li et al., 2024). This movement supports the deployment of Text-to-SQL systems in resource-constrained or privacy-sensitive settings by utilizing incremental pre-training and fine-tuning on SQL-centric corpora.
Test-Time Scaling and Ensemble Strategies	12/50 Papers	Emerging techniques such as self-consistency, ensemble voting, and tournament selection are being used to scale test-time computation and improve output quality (Yang et al., 2026) (Pengfei et al., 2025) (Liu et al., 2025) (BoYan et al., 2025) (Lei, 2025) (Lee et al., 2025) (Shen & Kejriwal, 2024) (Granado et al., 2025) (Gaetano et al., 2025) (Fan et al., 2024) (Li et al., 2024) (Tsatsi & Τσάτση, 2025). These strategies enhance robustness by generating multiple candidates and selecting the optimal query through verification-based heuristics, though they often introduce trade-offs in inference latency.

Chronological Review of Literature

The field of Text-to-SQL has transitioned from foundational neural and rule-based approaches toward sophisticated, agentic frameworks that prioritize system-level reliability and complex reasoning. Early research focused on leveraging large language models through prompt engineering and basic fine-tuning to improve execution accuracy on standard benchmarks. Recent advancements have shifted toward software-engineering-inspired architectures, multi-agent collaboration, and reinforcement learning to address the challenges of large-scale enterprise databases and semantic drift.

Year Range	Research Direction	Description
2024–2024	LLM Integration and Prompting	Research focused on harnessing proprietary and open-source LLMs through prompt engineering and initial fine-tuning to improve performance on standard benchmarks.
2025–2025	Agentic Frameworks and Reinforcement Learning	Methodologies evolved toward multi-agent systems, reinforcement learning, and test-time scaling to enhance reasoning, schema linking, and self-correction capabilities.
2026–2026	Enterprise-Grade Orchestration	The focus shifted to software-engineering-inspired workflows, human-in-the-loop verification, and iterative reasoning to solve complex, multi-turn queries in industrial environments.

Agreement and Divergence Across Studies

Current research demonstrates a strong consensus that agentic frameworks and iterative, execution-guided feedback loops significantly outperform traditional monolithic, single-pass Text-to-SQL architectures. While studies consistently validate that these methodologies improve performance on complex benchmarks like BIRD and Spider, they diverge on the optimal balance between model parameter size and architectural complexity. Discrepancies in reported accuracy and efficiency often stem from variations in the integration of specialized modules, such as schema retrieval, multi-agent collaboration, and the specific use of reinforcement learning versus prompt-based refinement.

Comparison Criterion	Studies in Agreement	Studies in Divergence	Potential Explanations
Architectural Paradigm	Agentic, multi-step frameworks are superior to monolithic models (Yang et al., 2026) (Hua et al., 2026) (Siobhan & Yassine, 2025) (BoYan et al., 2025) (Haolin et al., 2025).	Some studies prioritize streamlined, single-stage fine-tuning for efficiency (Song et al., 2025).	Differences in target deployment environments (e.g., enterprise vs. resource-constrained).
Schema Linking	Robust schema linking is critical for performance in large-scale databases (Duan et al., 2025) (Wang et al., 2025) (Duan et al., 2025) (Chen et al., 2025).	Approaches range from graph-based partitioning (Kiet et al., 2025) to vector-based retrieval (Sheng-min et al., 2025) and tool-assisted detection (Wang et al., 2025).	Variation in database scale and the complexity of semantic relationships within the schemas.
Test-Time Scaling	Scaling inference via ensemble voting or iterative refinement improves accuracy (Yang et al., 2026) (Pengfei et al., 2025) (Lei, 2025) (Gaetano et al., 2025).	Divergence exists between using Outcome Reward Models (ORMs) (Gaetano et al., 2025) versus majority voting (Yang et al., 2026) or tournament selection (Pengfei et al., 2025).	Differences in computational budget and the specific reward signals used for optimization.
Model Size vs. Performance	Smaller, fine-tuned models can match or exceed the performance of larger proprietary LLMs (Hua et al., 2026) (Kiet et al., 2025) (Sheng-min et al., 2025) (BoYan et al., 2025) (Li et al., 2024).	Some frameworks rely on the intrinsic reasoning capabilities of hyper-scaled models (Lei et al., 2024).	Discrepancies arise from the use of specialized training corpora and domain-specific fine-tuning.

Theoretical and Practical Implications

Theoretical Implications

- The transition from monolithic architectures to agentic, multi-expert frameworks challenges the traditional view of Text-to-SQL as a simple sequence-to-sequence translation task, suggesting instead that it is a complex software engineering problem requiring structured orchestration (Siobhan & Yassine, 2025) (BoYan et al., 2025) (Haolin et al., 2025).
- Evidence supports the theory that iterative, environment-driven feedback loops—such as execution-guided refinement and reinforcement learning—are essential for bridging the gap between syntactic generation and semantic correctness, which single-pass models fail to address (Hua et al., 2026) (Taicheng et al., 2025) (Lei, 2025).
- The findings suggest that test-time scaling, through strategies like ensemble voting and tournament selection, provides a viable mechanism for enhancing model reliability, effectively shifting the focus from model size to the depth of reasoning and exploration (Pengfei et al., 2025) (Granado et al., 2025) (Gaetano et al., 2025).

- Research indicates that schema linking is a distinct, critical bottleneck in large-scale databases, where semantic drift and ambiguity necessitate specialized retrieval and graph-based representation techniques rather than relying solely on the inherent reasoning capabilities of large language models (Duan et al., 2025) (Wang et al., 2025) (Duan et al., 2025).
- The emergence of frameworks that treat SQL generation as a multi-stage process—such as decomposing tasks into intermediate representations like Execution Description Language (EDL)—validates the theory that decoupling natural language understanding from formal query construction improves robustness in complex domains (Duan et al., 2025) (Duan et al., 2025) (Fan et al., 2024).

Practical Implications

- Industry practitioners should prioritize the adoption of agentic, modular frameworks over monolithic models, as these architectures demonstrate superior performance in handling the complex, multi-table queries typical of enterprise environments (Yang et al., 2026) (BoYan et al., 2025) (Eckmann et al., 2026).
- Organizations can achieve state-of-the-art performance using smaller, open-source language models by employing supervised fine-tuning and execution-guided reinforcement learning, thereby mitigating the privacy risks and high costs associated with proprietary, large-scale models (Sheng-min et al., 2025) (Mohammadreza, 2024) (Li et al., 2024).
- Policy and development standards for database interfaces should incorporate human-in-the-loop verification mechanisms, as these are proven to effectively address ambiguity and semantic mismatch in real-world scenarios where automated systems may still falter (Chen et al., 2025) (Eckmann et al., 2026).
- Research efforts should shift toward the development of dynamic, interaction-oriented benchmarks, such as those simulating evolving user intents, to better reflect the requirements of real-world business analytics and database exploration (Tianyu et al., 2025) (Lei et al., 2024) (Chen et al., 2025).
- For professional decision-making, the integration of outcome reward models and confidence-aware selection strategies offers a practical path to deploying reliable SQL generation systems that can autonomously abstain or request human intervention when confidence is low (BoYan et al., 2025) (Chen et al., 2025) (Gaetano et al., 2025).

Limitations of the Literature

Area of Limitation	Description of Limitation	Papers which have limitation
Semantic Mismatch	Models often fail to align natural language intent with database schemas, leading to semantic drift. This reduces the reliability of generated queries in large-scale databases where similar attributes cause confusion.	(Yang et al., 2026) (Duan et al., 2025) (Zhao et al., 2025) (Duan et al., 2025) (Tsatsi & Τσάτση, 2025)
Complex Reasoning	Current systems struggle with multi-table joins, nested queries, and domain-specific business logic. This limitation hinders the generalizability of models to real-world enterprise environments that exceed the complexity of standard benchmarks.	(Heng et al., 2025) (BoYan et al., 2025) (Lei et al., 2024) (Eckmann et al., 2026) (Gaetano et al., 2025)
Schema Linking Robustness	Inaccurate identification of database elements remains a critical failure point. Weaknesses in schema linking undermine the validity of the entire SQL generation process, especially when schemas are noisy or large.	(Yang et al., 2026) (Song et al., 2025) (Wang et al., 2025) (Chen et al., 2025)
Evaluation Gap	Existing benchmarks often fail to reflect real-world interactive scenarios or dynamic user intents. This creates a disconnect between high performance on static datasets and the actual utility of systems in practical applications.	(Hua et al., 2026) (Tianyu et al., 2025) (Mohammadreza, 2024) (Singh et al., 2024) (Lei et al., 2024)
Resource Constraints	High computational costs and reliance on proprietary models limit deployment feasibility. This restricts the accessibility of state-of-the-art Text-to-SQL solutions for privacy-sensitive or resource-constrained industrial settings.	(Sheng-min et al., 2025) (BoYan et al., 2025) (Mohammadreza, 2024) (Narodytska & Vargaftik, 2024) (Li et al., 2024) (Li et al., 2024)

Gaps and Future Research Directions

Gap Area	Description	Future Research Directions	Justification	Research Priority
Semantic Mismatch in Large-Scale Schemas	Current models struggle with semantic drift and schema ambiguity in databases with thousands of columns, leading to incorrect column selection (Duan et al., 2025) (Duan et al., 2025) (Tsatsi & Τσάτση, 2025).	Develop hierarchical schema-to-graph embedding models that incorporate domain-specific business logic to constrain the search space during retrieval.	Addressing semantic drift is critical for enterprise reliability, as current methods often fail to distinguish between semantically similar attributes (Duan et al., 2025) (Duan et al., 2025).	High
Real-World Enterprise Complexity	Existing benchmarks often fail to capture the complexity of multi-table joins and long-horizon workflows found in actual enterprise environments (Lei et al., 2024) (Eckmann et al., 2026).	Create synthetic benchmark generation pipelines that simulate multi-turn, multi-table enterprise workflows exceeding 100 lines of code.	Bridging the gap between static benchmarks and real-world enterprise SQL is essential for practical deployment and system-level reliability (Lei et al., 2024) (Eckmann et al., 2026).	High
Computational Latency in Agentic Frameworks	Iterative refinement, multi-agent collaboration, and test-time scaling significantly increase inference latency, hindering real-time applications (Yang et al., 2026) (Kiet et al., 2025) (Taicheng et al., 2025).	Investigate knowledge distillation techniques to compress multi-agent reasoning trajectories into single-pass, high-performance student models.	Reducing latency is necessary for interactive database exploration where users require immediate feedback during multi-turn sessions (Taicheng et al., 2025) (Ghosh et al., 2025).	Medium
Robustness to Adversarial Perturbations	Many state-of-the-art models exhibit significant performance drops when faced with adversarial perturbations or synonym-based schema changes (Liu et al., 2024) (Li et al., 2024).	Implement robust training regimes using adversarial data augmentation and confusion-aware sampling to improve model stability across SQL dialects.	Ensuring robustness against noisy schema information is vital for maintaining system reliability in unpredictable, real-world database settings (Song et al., 2025) (Liu et al., 2024).	High
Human-in-the-Loop Integration	While human-in-the-loop mechanisms improve accuracy, there is a lack of standardized protocols for when and how to solicit human feedback efficiently (Chen et al.,	Design adaptive, uncertainty-aware triggers that automatically identify high-ambiguity queries and request human	Minimizing human effort while maximizing accuracy is key to scaling Text-to-SQL systems in resource-constrained enterprise environments (Chen et al.,	Medium

Gap Area	Description	Future Research Directions	Justification	Research Priority
	2025) (Eckmann et al., 2026).	intervention only when necessary.	2025) (Eckmann et al., 2026).	
Reinforcement Learning Reward Signal Quality	RL-based training is sensitive to reward signal quality, and current methods struggle to distinguish between correct and erroneous SQL in self-play (Yuhao et al., 2025) (Ghosh et al., 2025).	Develop outcome reward models that utilize execution-guided feedback and semantic similarity metrics to provide more granular, reliable training signals.	Improving reward signal quality is essential for aligning model outputs with user intent and reducing the reliance on massive training datasets (Yuhao et al., 2025) (Kattamuri et al., 2025) (Gaetano et al., 2025).	Medium
Generalization Across SQL Dialects	Models often show performance degradation when moving between different SQL dialects or when training data for specific dialects is limited (Heng et al., 2025) (Kattamuri et al., 2025).	Explore cross-dialect transfer learning frameworks that leverage shared structural representations to improve performance in low-resource dialect settings.	Enhancing cross-dialect generalization is necessary for deploying Text-to-SQL systems across diverse, heterogeneous database infrastructures (Heng et al., 2025).	Medium
Interpretability of Reasoning Steps	Current agentic frameworks often act as "black boxes," making it difficult for users to understand why a specific SQL query was generated (Siobhan & Yassine, 2025) (BoYan et al., 2025).	Integrate explainable AI (XAI) modules that visualize the intermediate reasoning steps and schema linking decisions made by the agentic framework.	Transparency is crucial for building user trust and facilitating the debugging of complex SQL queries in enterprise settings (Siobhan & Yassine, 2025) (BoYan et al., 2025).	Low
Lifelong Learning and Knowledge Maintenance	Current systems lack mechanisms to update their knowledge base as database schemas and business logic evolve over time (Chen et al., 2025).	Propose lifelong learning architectures that continuously update internal knowledge bases through database profiling and automated rule mining.	Maintaining system accuracy in dynamic enterprise environments requires models that can adapt to evolving data structures without catastrophic forgetting (Chen et al., 2025).	Medium

Overall Synthesis and Conclusion

The field of Text-to-SQL has undergone a fundamental paradigm shift, transitioning from monolithic, single-pass architectures to sophisticated, agentic frameworks that treat database interaction as a structured software engineering problem. Current research establishes that while large language models provide a powerful foundation for natural language understanding, their reliance on intrinsic reasoning is insufficient for the complexities of enterprise-scale databases. Instead, the state of the art is defined by iterative, environment-driven feedback loops, multi-agent collaboration, and test-time scaling strategies that prioritize reliability and semantic correctness over simple syntactic generation.

Methodologically, the literature highlights that schema linking remains the most critical bottleneck in achieving robust performance. Advanced techniques, such as graph-based schema representation, cluster-based retrieval, and tool-assisted verification, have proven essential for mitigating semantic drift and ambiguity in large-scale environments. Furthermore, the integration of reinforcement learning—particularly through execution-guided feedback and outcome reward models—has emerged as a superior approach for aligning generated queries with actual database states. These methodologies allow smaller, fine-tuned open-source models to achieve performance parity with, or even outperform, massive proprietary systems, offering significant advantages in terms of privacy, cost, and deployment efficiency.

Despite these advancements, a persistent gap remains between performance on static, curated benchmarks and the unpredictable, noisy nature of real-world enterprise workflows. Current systems often struggle with long-horizon reasoning, multi-table joins, and evolving user intents that require dynamic interaction. The literature suggests that future research must prioritize the development of lifelong learning architectures capable of maintaining knowledge bases as business logic evolves. Additionally, there is a clear need for standardized protocols regarding human-in-the-loop verification, as autonomous systems still require human intervention to resolve high-ambiguity queries.

In conclusion, the transition toward agentic, modular frameworks represents a significant maturation of the field. While current methodologies have successfully addressed many syntactic challenges, the principal unresolved issues involve scaling these systems to handle massive, heterogeneous schemas and ensuring interpretability in complex reasoning trajectories. Future efforts should focus on bridging the gap between automated generation and human-aligned, reliable database exploration to fully realize the potential of Text-to-SQL in industrial applications.

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